

Smart Device Behavioral Analysis for Women's Health Tech

Prepared for: Executive Leadership and Marketing Team at Bellabeat

Project: Google Data Analytics Capstone Case Study — Smart Device Usage Habits Analysis

Tools Used: Google BigQuery (SQL), Google Sheets (Statistical Analysis), and Looker Studio

1. Executive Summary

This study analyzes the activity, sedentary behavior, and sleep data of smart fitness tracker users to identify key behavioral patterns. Through processing and statistical evaluation of the data, it was determined that walking is the primary driver of daily caloric expenditure (explaining nearly 90% of the variance in energy consumption) and that more than half of the female users exhibit activity levels below international health recommendations.

Based on these findings, a strategy is proposed targeting feature development across the Bellabeat App and devices (Leaf, Time, Ivy), focusing on gamified walking incentives, smart sedentary reminders, and specific challenges to overcome lower physical performance days.

2. Phase 1: Ask

Context and Business Problem

Bellabeat is a high-tech, women-oriented health company that manufactures smart fitness tracking devices. To position itself as a key player in the global wellness market, executive management needs to understand how consumers use non-Bellabeat smart devices (Fitbit data) and apply these insights to the brand's marketing and product strategy.

Key Business Tasks

- What are some of the trends in smart device usage?
- How could these trends apply to Bellabeat customers?
- How can these trends help influence Bellabeat marketing strategy?

3. Phase 2 & 3: Prepare & Process

Data Selection

The open-source **FitBit Fitness Tracker Data** (via Kaggle) was used. Two primary files were specifically selected:

- **dailyActivity_merged.csv**: Provides a comprehensive view of steps, distance by intensity, active minutes, caloric expenditure, and sedentary time.
- **minuteSleep_merged.csv / sleepDay_merged.csv**: Provides night-time rest monitoring and time spent in bed.

Selection Rationale: Files with insufficient sample sizes or excessive granularity lacking daily impact were discarded, prioritizing the completeness of daily sampling across 34 unique users.

Loading and Cleaning in BigQuery

To prevent schema update conversion errors (**Invalid schema update / Invalid time zone**), an **External Table** was used to process text strings (**STRING**) and parse date formats correctly.

SQL Query

```
-- Step 1: Create External Table for sleep data ingestion
CREATE OR REPLACE EXTERNAL TABLE
`project-941e89dd-df86-4a8c-ba4.Bellabeat.sleep_day_raw` (
  Id INT64,
  date STRING,
  value INT64,
  logId INT64
)
OPTIONS (
  format = 'CSV',
  uris =
['gs://bucket-cyclistic/cyclistic_folder/Bellabeat/minuteSleep_merged.csv'],
  skip_leading_rows = 1,
  allow_quoted_newlines = TRUE
);

-- Step 2: Unify and Clean Data in bellabeat_daily_clean
CREATE OR REPLACE TABLE
`project-941e89dd-df86-4a8c-ba4.Bellabeat.bellabeat_daily_clean`
AS
WITH sleep_clean AS (
  SELECT
```

```

        CAST(Id AS STRING) AS user_id,
        EXTRACT(DATE FROM PARSE_TIMESTAMP('%m/%d/%Y %I:%M:%S %p',
date)) AS activity_date,
        COUNT(date) AS total_time_in_bed,
        COUNTIF(value IN (1, 2)) AS total_minutes_asleep,
        COUNTIF(value = 3) AS minutes_awake_in_bed
    FROM `project-941e89dd-df86-4a8c-ba4.Bellabeat.sleep_day_raw`
    GROUP BY user_id, activity_date
),
activity_clean AS (
    SELECT
        CAST(Id AS STRING) AS user_id,
        CAST(ActivityDate AS DATE) AS activity_date,
        TotalSteps AS total_steps,
        TotalDistance AS total_distance,
        VeryActiveMinutes AS very_active_minutes,
        FairlyActiveMinutes AS fairly_active_minutes,
        LightlyActiveMinutes AS lightly_active_minutes,
        SedentaryMinutes AS sedentary_minutes,
        Calories AS calories
    FROM
`project-941e89dd-df86-4a8c-ba4.Bellabeat.daily_activity_raw`
)
SELECT
    a.user_id,
    a.activity_date,
    FORMAT_DATE('%A', a.activity_date) AS day_name,
    EXTRACT(DAYOFWEEK FROM a.activity_date) AS day_of_week,
    a.total_steps,
    a.total_distance,
    a.calories,
    a.very_active_minutes,
    a.fairly_active_minutes,
    a.lightly_active_minutes,
    (a.very_active_minutes + a.fairly_active_minutes +
a.lightly_active_minutes) AS total_active_minutes,
    a.sedentary_minutes,
    s.total_minutes_asleep,
    s.total_time_in_bed,
    IFNULL(s.minutes_awake_in_bed, 0) AS minutes_awake_in_bed
FROM activity_clean a

```

```
LEFT JOIN sleep_clean s
  ON a.user_id = s.user_id
  AND a.activity_date = s.activity_date
WHERE a.total_steps > 0
  AND a.calories > 0;
```

4. Phase 4: Analyze

Executed Queries and Obtained Results

Query 1: Overall Average Metrics

```
SELECT
  COUNT(DISTINCT user_id) AS total_unique_users,
  ROUND(AVG(total_steps), 0) AS avg_daily_steps,
  ROUND(AVG(calories), 0) AS avg_calories,
  ROUND(AVG(sedentary_minutes) / 60, 2) AS avg_sedentary_hours,
  ROUND(AVG(total_active_minutes), 0) AS avg_active_minutes,
  ROUND(AVG(total_minutes_asleep) / 60, 2) AS avg_sleep_hours
FROM
  `project-941e89dd-df86-4a8c-ba4.Bellabeat.bellabeat_daily_clean`;
```

Result: 34 users | 7,555 steps/day | 2,284 kcal/day | 15.72
sedentary hours | 229 active minutes | 7.06 sleep hours.

Query 2: Activity Trends by Day of the Week

```
SELECT
  day_of_week,
  day_name,
  COUNT(user_id) AS total_records,
  ROUND(AVG(total_steps), 0) AS avg_steps,
  ROUND(AVG(calories), 0) AS avg_calories,
  ROUND(AVG(sedentary_minutes) / 60, 1) AS avg_sedentary_hours,
  ROUND(AVG(total_minutes_asleep) / 60, 2) AS avg_sleep_hours
FROM `project-941e89dd-df86-4a8c-ba4.Bellabeat.bellabeat_daily_clean`
GROUP BY day_of_week, day_name
ORDER BY day_of_week;
```

Results:

day_of_wee	day_name	total_regi	avg_steps	avg_calori	avg_sedent	avg_sleep_
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k		stros		es	ary_hours	hours
1	Sunday	61	7150	2283	15,9	7,39
2	Monday	59	8204	2336	16,4	73
3	Tuesday	60	5980	1882	12,7	6,39
4	Wednesday	45	8011	2406	16,4	7,8
5	Thursday	42	7825	2359	16,7	6,41
6	Friday	64	7685	2351	16,8	6,85
7	Saturday	65	8181	2412	15,4	7,48

Statistical Evaluation of Steps vs. Calories Relationship

Based on these results, it was necessary to verify the relationship between steps and caloric expenditure. An analysis in Google Sheets on the steps and calories columns yielded the following statistical metrics:

- Pearson Correlation Coefficient (r): 0.946 (Very strong positive correlation).
- Coefficient of Determination (R²): 0.896 (89.6%).
- Interpretation: 89.6% of the variance in daily caloric expenditure is explained by a direct linear relationship with total step count. This quantitatively demonstrates that the "Tuesday Slump" (drop to 5,980 steps and 1,882 kcal) is driven by the lack of active walking on that day.

Query 3: Activity Level Segmentation (CDC / WHO)

```
WITH user_averages AS (
  SELECT
    user_id,
    AVG(total_steps) AS avg_steps
  FROM
    `project-941e89dd-df86-4a8c-ba4.Bellabeat.bellabeat_daily_clean`
  GROUP BY user_id
)
SELECT
  CASE
    WHEN avg_steps < 5000 THEN 'Sedentary (< 5,000 steps)'
    WHEN avg_steps BETWEEN 5000 AND 7499 THEN 'Lightly Active (5,000 - 7,499)'
    WHEN avg_steps BETWEEN 7500 AND 9999 THEN 'Moderately Active (7,500 - 9,999)'
    ELSE 'Very Active (>= 10,000 steps)'
  END AS activity_category,
```

```

COUNT(user_id) AS user_count,
ROUND(COUNT(user_id) * 100.0 / SUM(COUNT(user_id))
OVER(), 1) AS percentage
FROM user_averages
GROUP BY activity_category
ORDER BY user_count DESC;

```

Results:

categoria_actividad	user_count	percentage
Sedentary (< 5,000 steps)	12	35.3
Very Active (>= 10,000 steps)	8	23.5
Moderately Active (7,500 - 9,999)	8	23.5
Lightly Active (5,000 - 7,499)	6	17.6

5. Phase 5: Share (Visualizations)

Below is a summary of the key graphical representations included in the report:

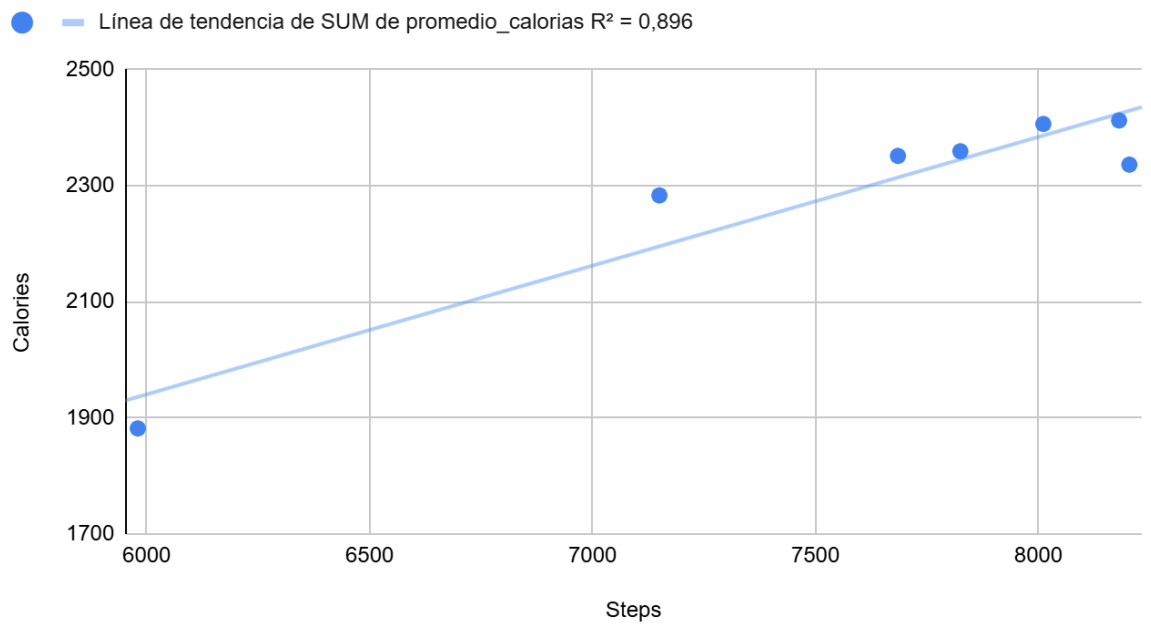
- **Dual-Y Combo Line Chart (Steps vs. Calories by Day):** Displays symmetry between both metrics throughout the week, visually highlighting Tuesday's drop and Saturday's peak.

Steps vs Calories by day



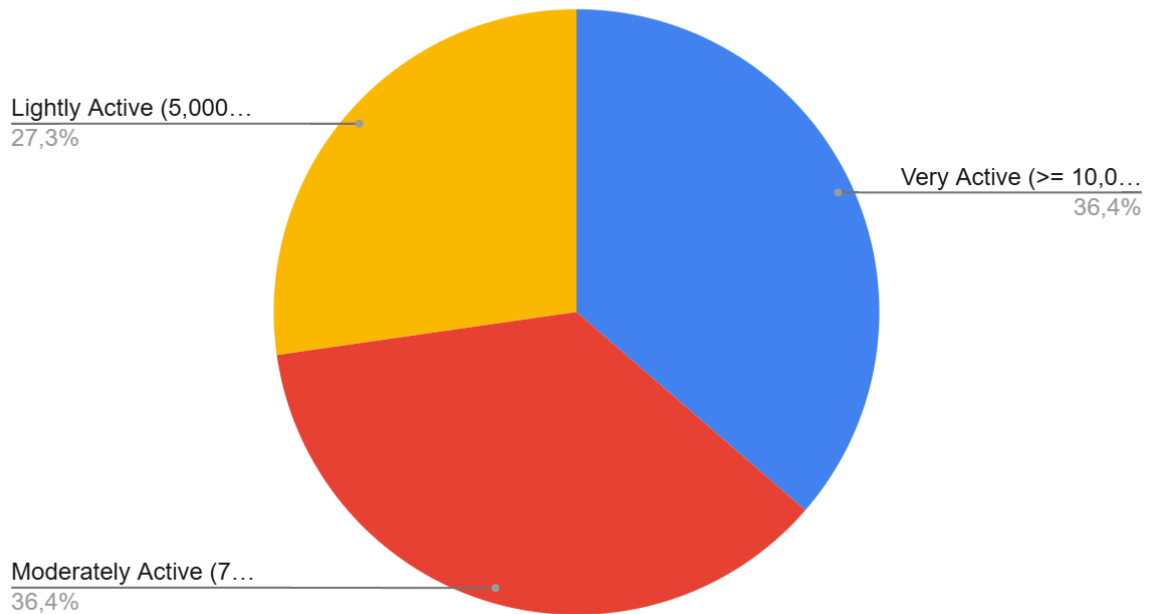
- **Scatter Plot and Regression:** Illustrates data concentration along the trendline ($R^2 = 0.896$), confirming the near-direct correlation ($r = 0.946$).

Steps vs Calories by day



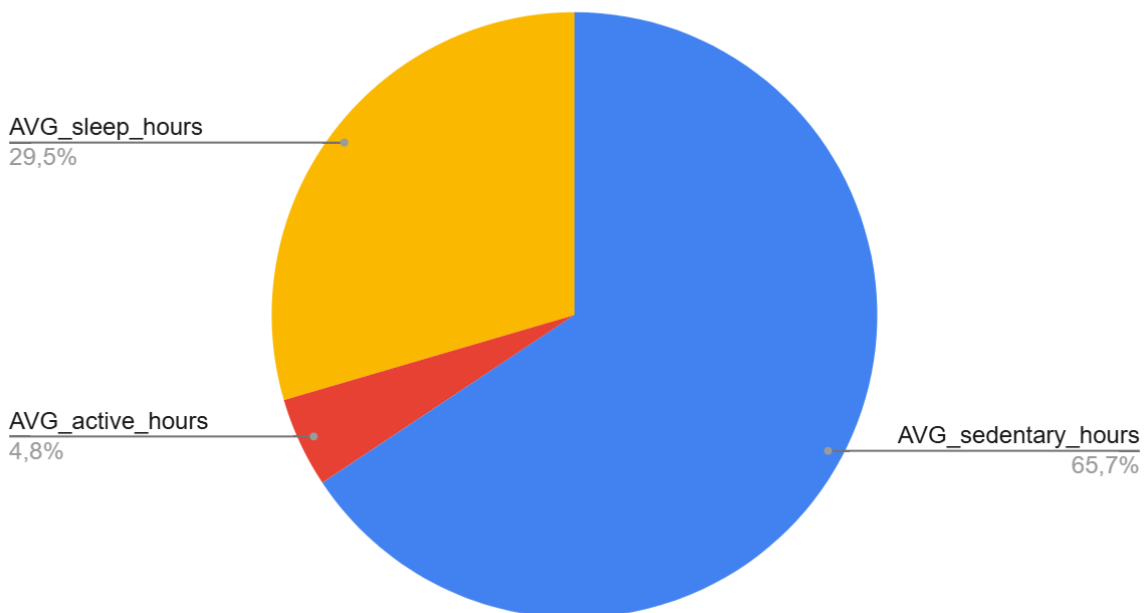
- **Donut Chart (User Segmentation):** Shows that 63.7% of users fall below the target of 7,500 steps per day, highlighting an opportunity for retention and education via the App.

Users Segmentation



- **Daily Time Composition:** A chart contrasting sedentary hours (65.5%) against active time (4.8%) and rest time (29.5%).

Daily Time Composition



6. Phase 6: Act (Business Recommendations)

Based on the mathematical and behavioral evidence gathered, three strategic recommendations are presented for the Bellabeat App and ecosystem devices:

1. Gamified Step Incentive Strategy

- **Rationale:** Caloric expenditure is 89.6% dependent on step count, and 35.3% of female users exhibit sedentary activity levels.
- **Action:** Implement a rewards system within the Bellabeat App where accumulating daily steps earns "Bellabeat Points". These points can be redeemed for free months of Bellabeat Premium or discounts on product purchases (Leaf, Time, Ivy).

2. Targeted Challenge Campaign to Overcome the "Tuesday Slump"

- **Rationale:** Tuesday records the lowest step average (5,980 steps) and caloric expenditure (1,882 kcal) of the entire week.
- **Action:** Design push notifications and double-point multipliers on Tuesdays and Thursdays in the App to motivate users to complete short 10-to-15-minute walks, stabilizing weekly energy consumption.

3. Smart Anti-Sedentary Alerts and Sleep Optimization

- **Rationale:** Users spend an average of 15.72 hours per day in a sedentary state.
- **Action:** Configure discreet reminders on Bellabeat devices upon detecting more than 50 minutes of continuous inactivity during work hours, prompting users to take 250 steps. Additionally, send wind-down alerts on days with lower average sleep duration (Tuesdays and Thursdays).